

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 1A

1. Special Signals

As it turns out, there are quite a few special signals in the world of signal processing. For each of the following, plot the signal on the domain $[-4, 4]$. Then, answer the following questions:

- i: Is the signal continuous or discrete?
- ii: Is the signal even? (i.e. $x(t) = x(-t) \forall t$, or $x[n] = x[-n] \forall n$)
- iii: Is the signal odd? (i.e. $x(t) = -x(-t) \forall t$, or $x[n] = -x[-n] \forall n$)

(a) Heaviside Step Function: $f : \mathbb{R} \rightarrow \mathbb{R}$, where $\forall t \in \mathbb{R}, f(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$

(b) Ramp Function: $f : \mathbb{Z} \rightarrow \mathbb{R}$, where $\forall n \in \mathbb{Z}, f[n] = \begin{cases} 0 & \text{if } n < 0 \\ n & \text{if } n \geq 0 \end{cases}$

(c) Rectangular Function: $f : \mathbb{R} \rightarrow \mathbb{R}$, where $\forall t \in \mathbb{R}, f(t) = \begin{cases} 1 & \text{if } |t| \leq 0.5 \\ 0 & \text{if } |t| > 0.5 \end{cases}$

(d) Kronecker Delta Function: $\delta : \mathbb{Z} \rightarrow \mathbb{R}$, where $\forall n \in \mathbb{Z}, \delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases}$

2. Interacting Discrete-Time Signals

Signals can be thought as vectors! Let's see how this is useful. Let the DT ramp signal $f[n]$ be defined as:

$$f[n] = \begin{cases} 0, & n < 0 \\ n, & n \geq 0 \end{cases}$$

Now define another signal:

$$g[n] = \begin{cases} 2, & n = -2, -1, 0, 1 \\ 0, & \text{otherwise} \end{cases}$$

For each question below we only consider the interval $-4 \leq n \leq 4$

(a) Sketch the signal $g[n]$

(b) Find and sketch a new signal $h[n] = f[n] + g[n]$

(c) Find and sketch another new signal $k[n] = 3f[n] - g[n]$

3. Inner Product Properties

For this question, we will verify our definition of the Euclidean inner product in Cartesian coordinates

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n, \quad \text{for any } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

indeed satisfies the key properties required for all inner products for the 2-dimensional case. Suppose $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^2$ for the following parts (there is a slightly different definition for complex vectors):

- (a) Show symmetry: $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$.

- (b) Show linearity: $\langle \mathbf{x}, c\mathbf{y} + d\mathbf{z} \rangle = c\langle \mathbf{x}, \mathbf{y} \rangle + d\langle \mathbf{x}, \mathbf{z} \rangle$, where $c, d \in \mathbb{R}$ are real numbers.

- (c) Show non-negativity: $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$, with equality if and only if $\mathbf{x} = \mathbf{0}$.